

paper

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2026-06-07

Introduction

- context for study established with – relevance to NCOS – relevance to other ecosystems
- clear connection between ideas presented in contextualization (i.e. main findings from other papers are synthesized, not just listed)
- articulated main question(s) for study

Methods

- description of temporal and spatial coverage of dataset
- references to monitoring report or any other NCOS-related publications outlining data collection process
- description of frequency of data collection
- description of statistical analysis or otherwise descriptive statistics (should be linked to question)

Results

Monthly precipitation at NCOS followed a strongly seasonal pattern across water years 2020–2025, with rainfall concentrated in winter months and dropping to near zero each summer (Figure 1). The highest recorded monthly precipitation occurred in early 2024, while intervening summers were consistently dry. This seasonal structure in precipitation provides the environmental backdrop against which waterfowl abundance patterns are interpreted.

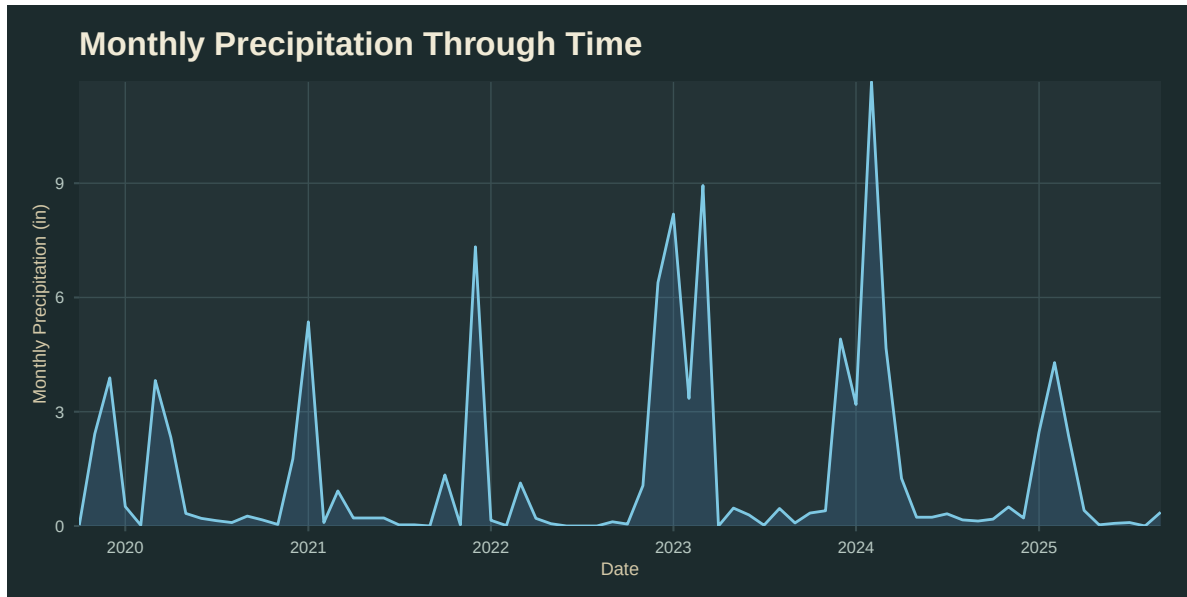


Figure 1: Monthly total precipitation (inches) at NCOS across water years 2020–2025.

Total monthly waterfowl abundance also peaked in winter months and was highest in early 2020, with a general decline in peak abundance observed across the study period (Figure 2). Mallard, Canada Goose, and Green-winged Teal were the dominant species across all years, together comprising the majority of monthly counts. Abundance dropped sharply each summer, with near-zero counts during June through August in most water years. Gaps in the figure reflect months with no survey conducted or no waterfowl observed.

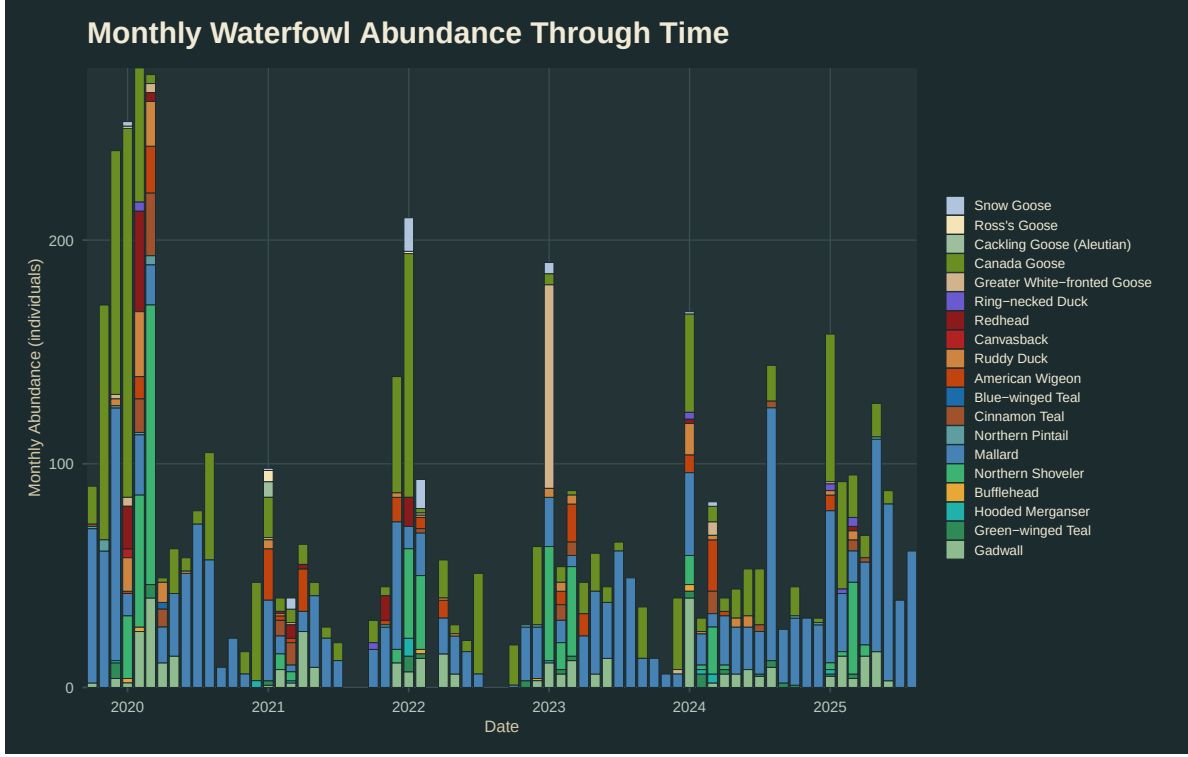


Figure 2: Monthly waterfowl abundance at NCOS across water years 2020–2025, stacked by species.

To address our primary research question, we plotted total monthly waterfowl abundance against the previous month's precipitation and fit a linear regression (Figure 3). Points were heavily clustered near zero precipitation with no clear directional trend, and the regression confirmed no statistically significant relationship between lagged precipitation and waterfowl abundance ($p = 0.085$, $R^2 = 0.046$). While the positive coefficient for lagged precipitation (5.61 individuals per inch) suggests a weak positive trend, it does not meet the conventional threshold for statistical significance ($\alpha = 0.05$), and the model explains less than 4% of the variation in monthly abundance (adjusted $R^2 = 0.031$).

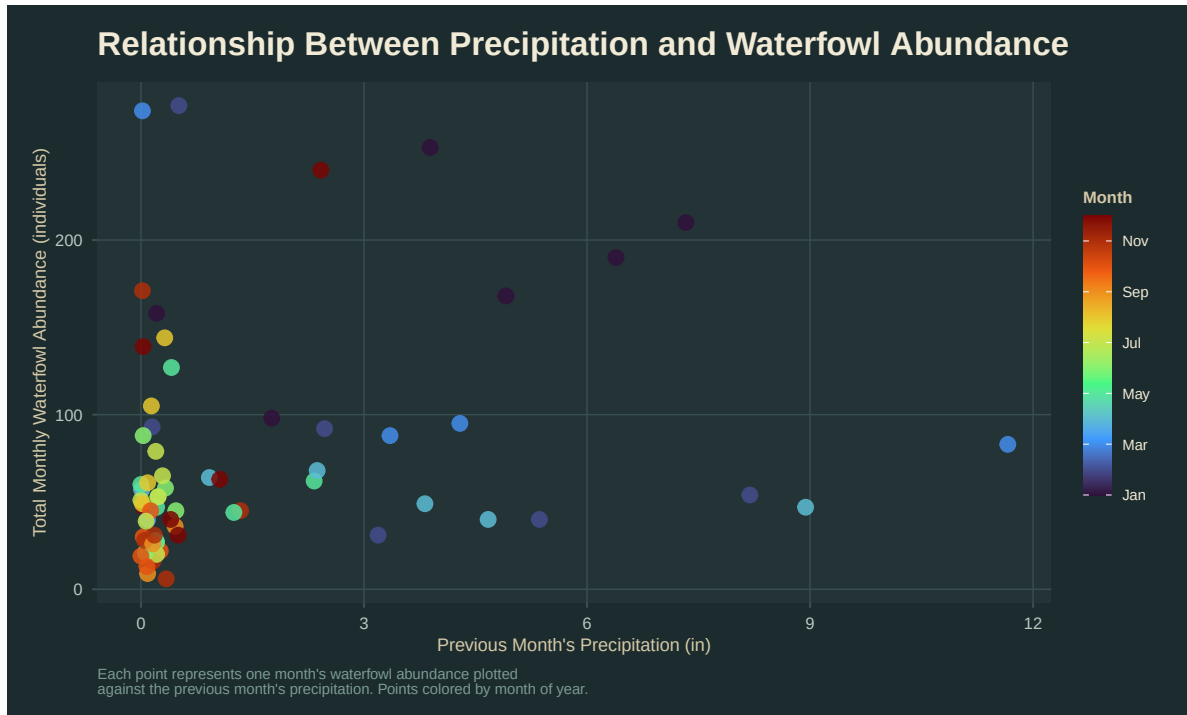


Figure 3: Total monthly waterfowl abundance plotted against the previous month's precipitation at NCOS across water years 2020–2025.

Despite the absence of a direct predictive relationship, Figure 4 reveals that waterfowl abundance and precipitation follow parallel seasonal trajectories when averaged across water years. Both variables peak in winter and decline steadily through spring, summer, and fall. This pattern suggests that while the amount of precipitation in a given month does not predict how many waterfowl will be present the following month, the two variables co-occur seasonally at NCOS, likely because both are driven by the same underlying seasonal forcing rather than one causing the other.

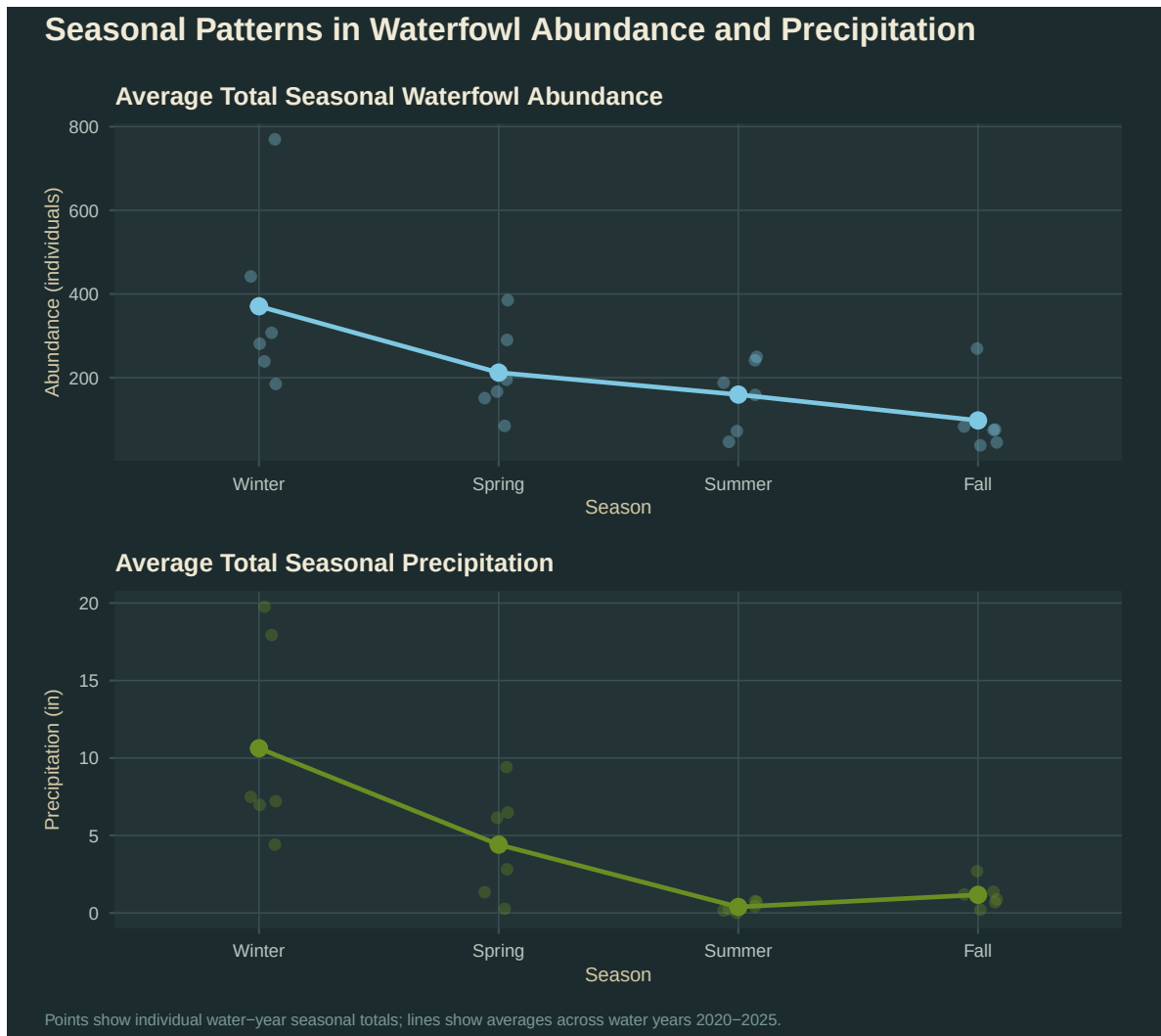


Figure 4: Average total seasonal waterfowl abundance (top) and precipitation (bottom) at NCOS across water years 2020–2025.

Discussion

- connecting results to NCOS reports or publications
- connecting results to other ecosystems
- articulation of challenges to collecting data, including: – what would be difficult about collecting the data you worked with – what other variables would influence the patterns you see in your chosen dataset, how those variables are collected, and why those variables

are relevant to your dataset – temporal changes that may influence patterns in your dataset, and how those changes are captured (or not) with the collection of your dataset